

nected to the Internet and to local-area networks, facilitating the spread of worms. However, with networks such as the World Wide Web in place, running services such as e-mail and file sharing systems such as P2P, viruses can spread seemingly without a host. This has led to a blurring of the line between viruses and worms.

Viruses can infect different types of hosts. The most common targets are executable files that contain application software or parts of the operating system. Viruses have also infected the executable boot sectors of floppy disks and script files of application programs. Additionally, viruses can infect files in other ways than simply inserting a copy of their code into the code of the host program. For example, a virus can overwrite its host with the virus code, or it can use a trick to ensure that the virus program is executed when the user wants to execute the host program. For a virus to replicate, it must be allowed to execute the code and write to memory. This is the prime reason why many viruses attach themselves to executable files that may be part of a legitimate program. When a user attempts to use the program, the code may be executed before the actual program runs.

“Elk Cloner”, a virus written by Rick Skrenta is recorded as the first computer virus to appear “in the wild” that is, outside the single computer or lab where it was created. Written in 1982, it attached itself to the Apple DOS 3.3 operating system and spread by floppy disk.

Before computer networks became widespread, most viruses spread on removable media, such as floppy disks. Some viruses spread by infecting programs stored on these disks, while others installed themselves into the disk boot sector, ensuring that they would be run when the user booted the computer from the disk.

As bulletin board systems and online software exchange became popular in the late 1980s and early 1990s, more viruses were written to infect popularly traded software.